

Internal hackathon

Supply Chain Forecasting

Coherent multi-level demand forecasting combining Temporal Fusion Transformers, Graph Neural Networks, and Optimal MinT Reconciliation.

PRESENTED BY

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CORE ARCHITECTURE

TFT + GNN + MinT Reconciliation

Hierarchical Discrepancies & Operational Impact

The Hierarchy Paradox

Independent forecasts at Store, RDC, and National levels produce conflicting totals:

National Forecast = 100,000 units

Sum of Regional Forecasts = 105,000 units

Discrepancy = 5,000 Phantom Units



Excess Inventory & Stockouts

Mismatched replenishment causing simultaneous overstock and stockouts.



Amplified Bullwhip Effect

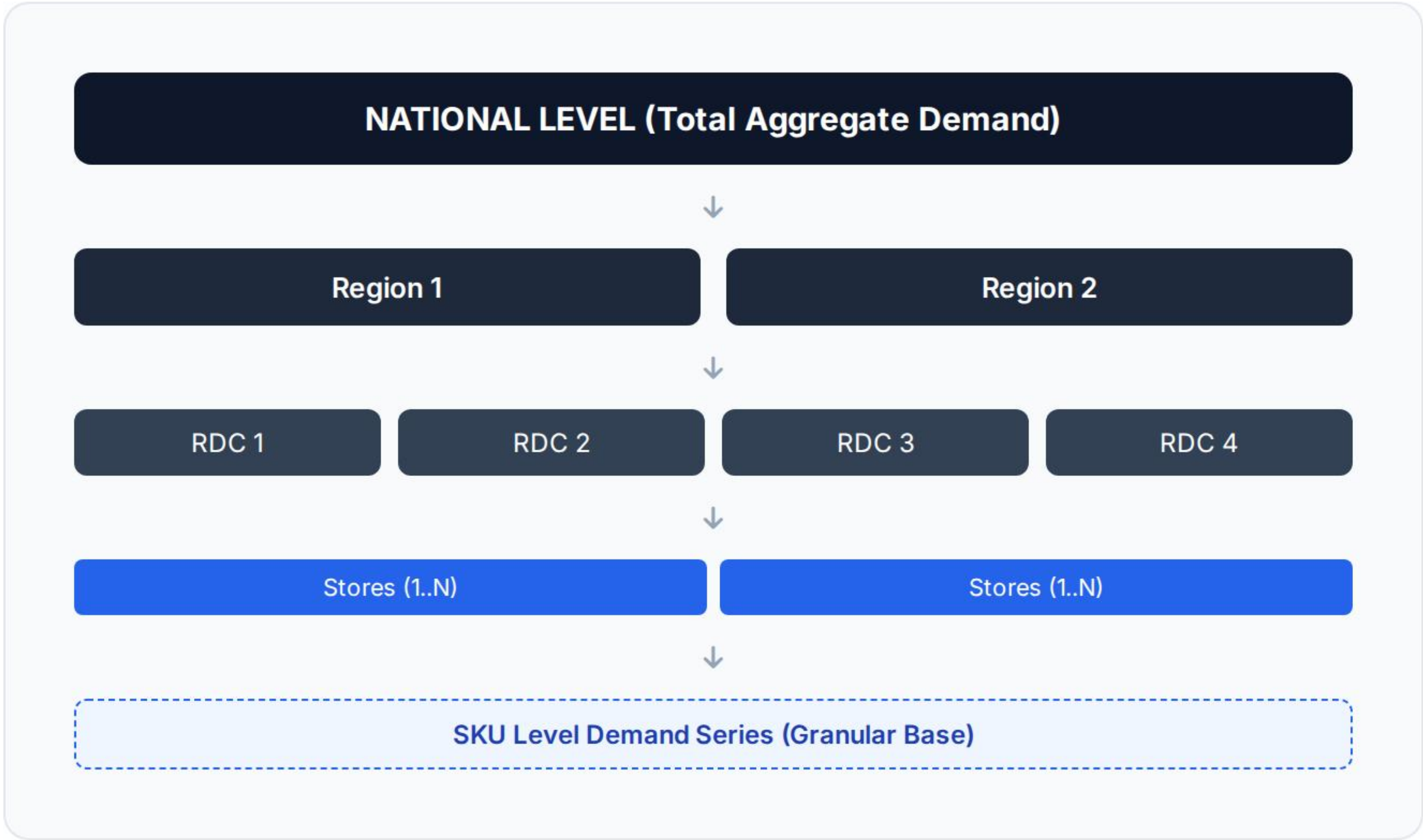
Small demand variance at stores distorts wholesale procurement upstream.



Increased Capital Holding Costs

Capital tied down in misallocated safety stock across distribution centers.

5-Level Supply Chain Aggregation Matrix



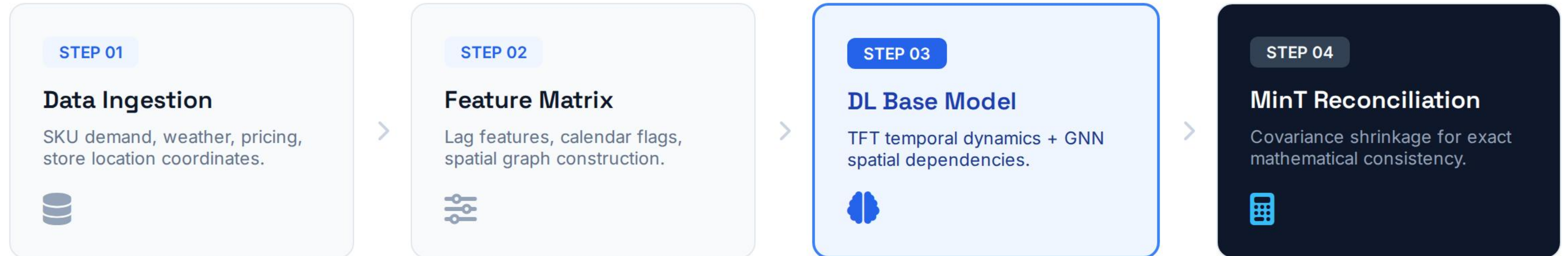
Mathematical Invariant

Bottom-up exact sum constraint: Every forecast at level L must sum precisely to parent nodes at level $L-1$.

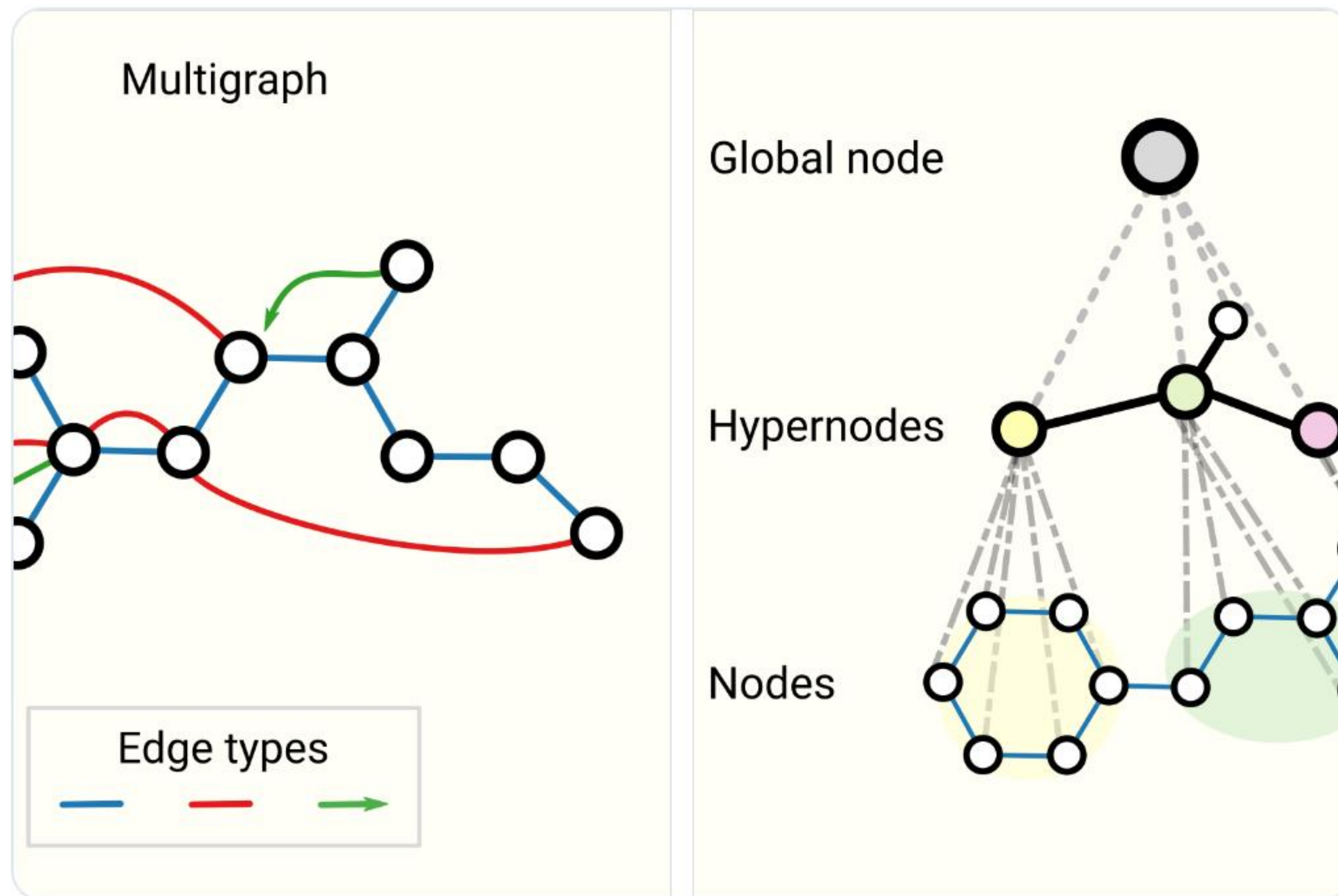
Coherent Planning

Eliminates manual overrides between procurement directors, regional managers, and store operators.

Pipeline Architecture



Deep Learning Architecture



1. Temporal Fusion Transformer (TFT)

Captures multi-horizon temporal dynamics, seasonal patterns, and static/time-varying covariates.

- Variable Selection Networks
- Gated Residual Network

2. Graph Neural Network (GNN)

Models spatial cross-store dependencies and geographic transfer effects across nodes.

- Spatial Distance Matrix
- Supply Chain Topology

Minimum Trace (MinT) Forecast Reconciliation

MINIMUM TRACE RECONCILIATION EQUATION

$$\tilde{Y} = SP\hat{Y}$$

$\hat{Y}$: Base Forecasts S : Summing Matrix P : Reconciliation Matrix $\tilde{Y}$: Coherent Output

Error Variance Minimization

Adjusts base predictions by minimizing total forecast error variance across all hierarchy levels.

Unbiased Adjustment

Preserves deep learning base accuracy while guaranteeing 100% aggregation consistency.

Feature Engineering Matrix



Temporal Signals

- Demand lags (t-1, t-7, t-30)
- Moving averages & std dev
- Day-of-week & seasonality



Spatial Data

- Store & RDC coordinates
- Geographic distances
- Transportation lead times



Commercial Drivers

- Product price & discounts
- Marketing promo flags
- Stockout indicators



External Exogenous

- Local temperature & rain
- Holiday/festival calendar
- Macroeconomic indicators

Multi-Horizon Decision Alignment

SHORT TERM

1–7 Days

Store Replenishment

Optimizes daily truck routing, store shelf dispatch, and immediate safety stock alerts.

MEDIUM TERM

14–28 Days

Distribution Planning

Balances inventory across regional distribution centers (RDCs) and warehouse labor shifts.

LONG TERM

1–3 Months

Procurement & Sourcing

Guides supplier contract negotiations, factory master production schedules, and raw material orders.

EXPERIMENTAL RESULTS

Progressive Model Performance (RMSSE Metric)

Model 1: Seasonal Naive Baseline	RMSSE: 1.00
Model 2: LightGBM / XGBoost ML	RMSSE: 0.82
Model 3: N-BEATS Deep Learning	RMSSE: 0.74
Model 4: Temporal Fusion Transformer (TFT)	RMSSE: 0.65
Model 5: TFT + Graph Neural Network (GNN)	RMSSE: 0.58
Model 6: TFT + GNN + MinT Reconciliation	RMSSE: 0.49 (Best)

Primary Metric

RMSSE

Root Mean Squared Scaled Error scale-independent metric for multi-level comparison.

Secondary Metrics:

- MAE & RMSE across SKUs
- WAPE & Forecast Bias

Business Benefits & Strategic Roadmap

Stakeholder Value

Demand Planners

Unified single-source forecast across all geographic zones.

Inventory Managers

Up to 22% reduction in safety stock holding costs.

Supply Chain Directors

Seamless procurement alignment and reduced bullwhip friction.

Future Scope

Probabilistic Intervals

Uncertainty estimation via quantile regression loss.

Digital Twin Integration

Direct ERP/WMS real-time API syncing for automated replenishment.

Dynamic Graph Topology

Automated spatial graph updates based on live traffic and disruptions.

SUMMARY

Coherent Demand Intelligence

By unifying Temporal Fusion Transformers, Graph Neural Networks, and MinT Reconciliation, our model provides mathematically consistent, highly accurate forecasts from SKU to National levels.

Thank You

Supply Chain Forecasting